

Not quite ‘out of the blue’?

Towards a dynamic, relevance-theoretic approach to thetic sentences in English

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1. Introduction

This paper outlines a relevance-theoretic approach to ‘thetic’ sentences (all-new, subject-stressed sentences) such as *my CAR broke down* or *JOHN arrived*, building on Sperber & Wilson (1995 – hereinafter S&W), Breheny (1996, 1998), and Sax (2010). It claims that a number of the distinctive traits of thetic sentences in English may be productively explained in terms of the relevance-theoretic comprehension procedure and the impact of early focal stress placement on dynamic processing. One conclusion is that the ‘out of the blue’ view of thetic sentences widely assumed in the information-structure literature is misleading; instead, a processing constraint introduced by what I call an ‘expectation for an explanation’ is found to be more explanatory.

Section 1 defines and encapsulates thetic sentences as the topic of the present study, exploring certain of their properties. Section 2 below lays out the types and functions of thetic sentences as proposed in the literature and considers the prominent ‘out of the blue’ analysis. Section 3 introduces a relevance-theoretic approach to dynamic processing and looks at its application to thetic sentences in greater depth. Section 4 then reexamines the properties, types, functions of thetic sentences in this light.

1.1 Thetic sentences – Preliminaries

Certain sentences in English (and other languages, see Sasse 2006), when pronounced with a stressed subject and unstressed predicate, can support an all-new reading, as in

the natural interpretation of (2) in context (1). Such sentences have long attracted attention in the information-structure literature (e.g. Schmerling 1976; Fuchs 1980; Selkirk 1984; Lambrecht 1994, 2000, among many others). Following influential work by Sasse (1987, 2006), here we will call these ‘thetic’ sentences¹ (or sentences that support ‘thetic’ readings), viewed in opposition to non-thetic or ‘categorical’ sentences (which are pronounced with focal stress on a predicate constituent, as well as possibly a secondary stress on the subject).

What is particularly interesting about (2), and thetic sentences in general, is when/why it is possible for a predicate to go unstressed, despite not being given in previous discourse:

- (1) context: employee X arrives late. Saying nothing, boss Y raises eyebrows and glances at clock to tacitly communicate to X: "Why are you so late?"
X responds...
- (2) My CAR broke down. (ŚP – thetic)²
- (3) # My CAR was booby-trapped. (ŚP – infelicitous attempt at thetic reading)

Much of the literature on thetic sentences is concerned with what Lambrecht (2000: 623) calls the ‘vexing’ question of why some predicates, such as (2), do allow thetic (all-new) readings in subject-stressed sentences, whereas others, such as (3), do not. It has widely been remarked that some close semantic relation must obtain between the subject and predicate for a given subject-stressed sentence to support a thetic reading. As S&W (1995: 211-212, from whom examples (2)-(4) have been drawn) write, in the case of (2)

¹ Some of the alternative labels that have been proposed in the literature are "news sentences" (Schmerling 1976), "all-new sentences" (Allerton & Cruttenden 1979, Fuchs 1980), and "sentence-focus constructions" (Lambrecht 1995, 2000).

² The schematic symbols S and P stand for subject and predicate, following the usage of Fuchs (1980). The conventional all-caps notation used here reflects certain gross simplifications of the facts of intonation, as follows: I assume that in every English sentence there is a single constituent (for our purposes either an NP, verb, or phrasal verb particle) that bears the most prominent nuclear pitch accent, which I call ‘focal stress’; this constituent I mark with full capitals and call the ‘(focally) stressed constituent’. The term ‘focal stress’, chosen in this paper for the sake of compatibility with S&W (1995) and Breheny (1996, 1998), is by and large analogous to Lambrecht’s (1994) ‘main sentence accent’ and Ladd’s (1996) ‘sentence stress’. Other pitch accents that precede the sentence stress I mark when appropriate with small-caps and call ‘secondary stress’. In a ‘subject-stressed’ sentence, the sentence stress falls on the subject NP while the predicate bears no nuclear pitch accent. Any unstressed constituent that follows the sentence stress is referred to as ‘post-focal’.

this is intuitively the fact that it is a widely-known, inherent property of cars that they sometimes break down, and that this sometimes causes unexpected delays. As evidence of this intuition, S&W note that altering the predicate to something more informative / less predictable as in (3) prevents a subject-stressed sentence from supporting a thetic reading and therefore (3) is inappropriate in response to situation (1). The categorical (4), by contrast, is felicitous in response to situation (1) precisely because it is **not** normally perceived to be an inherent property of cars that they might end up booby-trapped, or for this to be a plausibly expectable excuse for being late.

(4) My car was BOOBY-trapped. (ŚP – categorical)

As S&W (1995: 211-212) put it, at issue is whether in thetic sentences like (2) "stress has lost its natural highlighting function and become subject to arbitrary linguistic constraints, or whether its position can be seen as following in some interesting way from pragmatic principles". In what follows, I try to go some distance towards arguing the latter position, taking their lead.

1.2 'One fell swoop'

One insight widely discussed in the literature is that the semantic processing of thetic sentences seems to take place in a single step, rather than two steps in the case of categorical sentences. Thus whereas the processing of the categorical sentence in (6) below involves first identifying the referent of John, then predicating something of this referent (the event of his arrival), in the thetic (5) a situation is presented as a whole (Sasse 2006), whereby the "single accentuation signals communicative fusion of an element denoting an individual and an element denoting an event" (Sasse 1987: 522).

(5) JOHN arrived. (ŚP – thetic)
 (6) John ARRIVED. (ŚP – categorical)

Sasse's 'communicative fusion' bears strong affinities to what Chafe (1974) called the 'conceptual unity' between the subject and predicate in thetic sentences, or what Fuchs

(1980), Jacobs (1999), and others see as subject-predicate ‘integration’ for the purpose of semantic processing. Ultimately this presented-together-not-separately observation stretches back to the *thetic/categorical* terminological distinction as originally put forward by the 19th-century German philosophers Marty and Brentano – the linguistic term ‘*thetic*’ as used here being historically derived from their philosophical notions of a one-part ‘*thetic judgment*’ as opposed to a two-part ‘*categorical judgment*’.³

This semantic processing point is perhaps best summed up by Jacobs (1999: 57), referring to the *thetic* reading of a sentence like (7) below:

"We do not start semantic processing of this sentence by referring to a certain group of policemen, then go on by processing the meaning of [the predicate 'coming'] in a second step and finally combine the results of the two steps. Rather, [(7)] is semantically processed at one fell swoop, by referring holistically to an event of approaching policemen."⁴

(7) The POLICE are coming.

Intuitions on this point are thus widely noted; again the challenge as posed by S&W above is whether they may be seen to follow from more general (pragmatic) principles.

At this stage we should also briefly consider a certain complication to the neat *thetic* (ŚP) vs. *categorical* (SP) divide as presented in the previous section. In (8) below, both the subject and predicate are stress marked, thus licensing their introduction of a new referent and a new predicate, respectively. Example (8) might also be felicitous in responding to the context in (1) above, and thus might be seen as presenting 'all-new' information just as (2) does. However, I follow Lambrecht (1994, 2000) and others in assuming that ŚP sentences such as (8) are non-*thetic* by definition,⁵ and that such also involve processing in two steps, much like SP *categorical* sentences such as (6).

³ For more on the philosophical origins of the term "thetic" – Greek *θητικός*, from Ancient Greek *τίθημι* "put, pose" – see Sasse (1987, 2006); in this paper I will remain concerned with *thetic* utterances purely as a linguistic phenomenon.

⁴ Jacobs is in fact discussing the German equivalent of (7), "Die POLIZEI kommt", but his observation holds identically for the English translation shown here.

⁵ There is some confusion in the literature as to precisely how ŚP sentences should be classified along the divide we are exploring here. Schmerling, for instance, groups them together with *thetic* SP sentences like (2) into a broader class of 'news sentences.' Closer analysis of the separate processing impact of sentence (primary) stress and secondary stress in ŚP sentences like (8) unfortunately falls outside the scope of this paper; see for instance Büring (1999) on topic interpretation and nominal stress marking.

(8) My CAR broke DOWN. (Ś́ – categorical)

Pausing to summarize our discussion so far, we can say that thetic sentences, as they interest us in this paper, exhibit Ś́ stress marking and may be felicitous in discourse-initial contexts (even though the predicate is not given in previous discourse), present all-new information in ‘one fell swoop’, and exhibit a close relationship or communicative fusion between subject and predicate. They are viewed here in opposition to categorical sentences (which show Ś́ stress marking, are not felicitous in discourse-initial contexts, and predicate a new relation about a given subject) as well as in opposition to doubly-stressed Ś́ sentences (which may likewise present all-new information, yet do so in two parts); neither of the latter require/exhibit the same kind of close relation between subject and predicate.

1.3 ‘Out of the blue’

The acceptability of thetic sentences in discourse-initial position is considered one of their defining characteristics, and discourse-initial contexts such as (1) are widely used in the literature (e.g. Lambrecht 1994) as a diagnostic tool to test/force the possibility a thetic reading. A given subject-stressed sentence is held to have a thetic reading if and only if it can be construed as a felicitous response to an (explicit or implicit) question such as *What happened?*. In such contexts, chosen to ensure that neither the subject nor the predicate point back anaphorically to previous linguistic material, sentences are said to be “uttered out of the blue” (Selkirk 1984: 217).

Thetic sentences have attracted attention because the absence of stress on the predicate poses a distinct problem for bipartite approaches to information structure posited on the whole-sentence level – ‘bipartite’ here meaning all approaches which partition a complete sentence into two disjoint types of constituents: ‘old’/‘new’ information in various senses addressed by Prince (1981), ‘topic’/‘focus’ in information structure as presented by Lambrecht (1994) or by Erteschik-Shir (1997), or ‘theme’/‘rheme’ in the Prague school tradition (Firbas 1992). Because thetic sentences are acceptable in ‘out of the blue’ contexts, both the subject and predicate must

constitute new information (not anaphorically linked to previous linguistic material), yet the predicate receives no stress marking, thereby behaving as if it were old or given (anaphorically linked to previous linguistic material).

Sasse (1987) stakes the clear position that thethetic/categorical distinction is not a matter of information structure, as is often assumed, but can instead be explained in terms of ‘expectation’. We will have more to say about the notion of ‘expectation’ below; for now note that intuitively, although the predicate in athetic sentence is not given **before** the sentence is uttered, because of the close subject/predicate relationship the predicate might be seen as becoming given in some sense (or at least not entirely new) **while** the sentence is being uttered/processed.

2.2 Relationship to other focus readings

Lambrecht (1994, 2000), taking his cue from Construction Grammar, suggests that the subject-stressed form of thethetic sentence⁶ in (2), here repeated as (9), can be explained in terms of what he calls ‘paradigmatic contrast’ to the categorical sentence in (10):

(context: What happened?)

(9) [My CAR broke down_F]

(context: What happened to your car?)

(10) My car [broke DOWN_F]

Here in each example I have additionally marked the focus phrase, delimiting newly asserted material (following the practice in Lambrecht 1994; Erteschik-Shir 1997, etc.) Lambrecht sees the subject-stressed form of sentence (9) as a kind of ‘detopicalization’ strategy: signaling that the subject lies inside the focus phrase, rather than lying outside it as subjects customarily do and as the subject does in (10). Lambrecht (2000) maintains that thethetic reading of (9) arises because the subject looks and behaves in certain ways like an object – which in English means focal stress on the subject, no stress on the

⁶ For ‘thetic’ sentence Lambrecht uses the term ‘subject-focus construction’; for ‘categorical’ he uses the term ‘predicate focus construction’.

predicate – and that this is intended to maximally differentiate thethetic construction from the categorical construction. Space prevents a more detailed examination of Lambrecht's extensive cross-linguistic arguments here, but I will point out two shortcomings that arise in their application tothetic sentences in English.

Firstly, Lambrecht's notion of 'paradigmatic contrast' brings us no closer to explaining what he has called the 'vexing' question of why some predicates supportthetic readings in subject-stressed sentences, whereas others do not. Returning to S&W's example seen above as (3), repeated here as (11) with focus phrase marked, it remains to be explained why this sentence supports nothetic reading, even though its $\acute{S}P$ stress marking is likewise maximally distinct from categorical $\acute{S}P$ marking:

(context: What happened?)

(11) #[My CAR was booby-trapped_F]

For Lambrecht this is an issue to be deliberated within the class ofthetic sentences per se, being only indirectly traceable to the impact of subject stress marking, via the functional notion of 'paradigmatic contrast'. For Lambrecht, stress marking on the subject only serves to make that subject behave more like an object, thereby to differentiatethetic from categorical.

Secondly, another distinctive property ofthetic sentences (i.e. subject-stressed sentences which support athetic reading) is that in addition to what we have been calling athetic reading they are always compatible with another possible reading illustrated in (12), known in the literature as a 'narrow focus' or 'argument-focus' reading (e.g. Lambrecht 1994):

(context: What broke down?)

(12) [My CAR_F] broke down

The focus phrase in (12) is again intended to delimit the newly asserted material, here corresponding to the question word. Compare the narrow-focus (12) to the possibly phonologically identical⁷ example (9), repeated again, which bears athetic reading:

(context: What happened?)

(9) [My CAR broke down_F]

Comparing this minimal pair (9)/(12) turns up a different shortcoming of Lambrecht's analysis: a question of motivation. Under his concept of 'paradigmatic contrast' the purpose of $\acute{S}P$ stress marking onthetic utterances is to maximally differentiate them from the $S\acute{P}$ stress marking on categorical sentences. The fact that boththetic and narrow-focus readings of subject-stressed sentences share $\acute{S}P$ stress marking, on the other hand, for him apparently falls out as merely a coincidence, a by-product of how thethetic-categorical paradigmatic contrast is expressed in English. This raises the question of whether – contrary to Lambrecht's approach – there might indeed be some motivation for the similarity of the narrow-focus reading in (12) and thethetic reading in (9). I will return to this issue in Section 4.

1.4 Syntactic and lexical constraints

This subsection will briefly review some of the observed syntactic, lexical, and semantic constraints on which predicates may support athetic reading in a subject-stressed sentence. In terms of syntax, note that a large share of the core examples of thethetic phenomenon involve intransitive expressions such as (2) above. However,thetic readings are not limited to intransitive sentences – passives and some transitives, such as (13), are also possible:

(context: Why is SHE here?)

(13) Her HUSBAND beats her.

(Gussenhoven 1992)

⁷ On the issue of whether thethetic and narrow-focus readings may show / may not show / must show phonologically different realization of pitch accent, compare Ladd 1996; Selkirk 2002.

In lexical terms, a thetic reading is possible only when both subject and verb are fully lexical, i.e. do not involve pronouns or anaphoric predicates (Fuchs 1980: 454) – a sentence such as HE arrived supports only a narrow-focus reading, not a thetic reading. Objects, on the other hand, show the reverse tendency; if a transitive predicate supports a thetic reading there is an at least very strong preference for the object to be pronominal as in (13) above.⁸ A non-pronominal object, even apparently with the same referent, precludes a thetic reading in (14):

(context: Why is SHE here?)

(14) #Her HUSBAND beats the poor soul. (Gussenhoven 1992)

Moreover, Fuchs notes a strong tendency for the lexical material in a thetic sentence, especially the predicate, to be ‘centered’ to the default, current situation, which she explains as “oriented to the perspective established at the point of the communication”; moreover the choice of predicate itself is often deitically linked to current concerns: “especially so with verbs of (dis)appearance [...], presence, absence, and their causatives” (Fuchs 1980: 455-456), such as bring and take away.

Summing up various constraints on thetic readings, Fuchs argues that the underlying principle is a restriction on the amount of information conveyable, suggesting that deitic lexical items are easier to convey than non-deitic ones, pronominal objects easier to convey than lexical objects. Foreshadowing our own analysis below, we may venture to rephrase Fuchs' formulation as a cognitive restriction on the amount of information that the hearer can process in Jacobs' ‘one fell swoop’.

To recap, Section 1 has defined thetic sentences as the topic of this study, then reviewed characteristics and made observations which any approach to thetic sentences must account for sensibly. These include: why is it that a close semantic relation between subject and predicate should be required to support a thetic reading, whether there may be any motivation for the ambiguity observed by Lambrecht (1994) between the thetic and narrow-focus readings of subject-stressed sentences, and how/why constraints on the syntactic and lexical form of thetic sentences seem to limit the amount

⁸ The very possibility of object pronouns in thetic readings is another argument against the pure “out of the blue” approach to thetic sentences, but will not be pursued here.

and kind of information that may be processed in ‘one fell swoop’. These issues will all be readdressed in Section 4; in the meantime I will look at the taxonomy of types of thetic utterances in Section 2 and outline a dynamic approach to dynamic processing in Section 3.

2. Types and functions

This section provides an overview of various classifications of thetic sentences, synthesizing the taxonomies of several authors and seeking a common denominator in light of the above issues. This will pave the way for laying the foundations of a dynamic approach to the two communicative functions of thetic sentences, in terms of an expectation for an explanation.

2.1 Classification of types

Allerton & Cruttenden (1979 – hereinafter A&C) list three types of thetic sentences; the first consists of thetic sentences involving what they call ‘empty’ verbs, i.e. predicates which exhibit a particularly tight lexical semantic relationship with the subject NPs, as in (15) - (16):

(context: What's that noise?)

(15) The TELEPHONE's ringing.

(16) The KETTLE's boiling. (both A&C 1979: 52)

Pointing out the tight relationship in examples such as these, A&C note that a kettle is capable of few activities other than boiling, etc. But note that this relationship is further mediated by context – Fuchs notes that A&C's observations may only hold in the context of ‘household’ or other ‘participant-oriented’ issues (Fuchs 1980: 457).

A&C's second type of thetic sentences involves ‘verbs of (dis)appearance’ in which the subject NP “becomes conspicuous either by its sudden presence or by its absence”; appearance is illustrated in (17) - (18), disappearance in (19) - (20):

(context: What happened / What's going on?)

(17) JOHN arrived.

(18) The POLICE are coming.

(19) My KEY's disappeared. (A&C 1979: 52)

(20) The DOG's escaped. (A&C 1979: 52)

A&C's third and last group deals with events of 'misfortune', commonly involving damage and/or an accident befalling the subject, exhibited in (21) - (22) (and likewise seen in (1) above):

(context: What happened? / What's wrong?)

(21) The BRAKES failed.

(22) My SISTER died. (both A&C 1979: 52)

However, A&C's 'misfortune' category has been criticized by numerous authors, and expanded to include fortunate events as well, if the context supports them:

(context: Why are you so happy?)

(23) JOHN survived. (Fuchs 1980: 458)

(24) My CACTUS flowered. (Sasse 1987: 526)

Sasse (1987) suggests this third class of misfortune/fortune is more aptly recast as 'explanations'. Sasse also goes on to add a few further classes, including expressions involving body parts:

(context: What's wrong?)

(25) My TOOTH hurts.

(26) My ARM fell asleep. (both Sasse 1987: 521)

...weather expressions:

(context: What is the situation?)

(27) The SUN is shining.

(28) NIGHT is falling.

(both Sasse 1987: 525)

...and background, scene-setting descriptions:

(context: What was going on?)

(29) SILENCE descended.

(30) A BLIZZARD was approaching.

(both Sasse 2006: 286)

A&C (1979: 52) admit considerable overlap between their three categories, suggesting that they may ultimately be reducible to one notion in that they all “seem to draw the listener's attention to a prominent NP, saying 'look at NP'” which means “the subject is more 'newsworthy' [...] than the verb in the context where it appears”. Sasse (2006: 526), too, seeks a unifying factor, noting a common thread underlying his expanded set of classes in that they all “lead back to the pragmatic criterion of expectation” on the part of the hearer.

2.2 Communicative functions

Sasse (1987) and Lambrecht (1994) both point to two fundamental communicative functions played bythetic sentences, spanning across the above types. One function is primarily to introduce a new referent into the mutual cognitive environment; this is perhaps best exemplified by (17) and (18) above, which achieve little more than introducing such a new referent (the chief effect might be paraphrased as ‘John is / some police officers are now in the mutual cognitive environment’). I will follow Sasse in calling these ‘entity-central’ thetic sentences (an alternate term is Lambrecht's ‘presentational’ sentences).

The second communicative function is “to announce an event involving a new discourse referent” (Lambrecht 1994: 14). This function is perhaps best exemplified above by (15) and (16). In these two examples the communicative message does not rest mainly in pointing out the presence/existence of the referent of the subject NP (i.e. intuitively the message is **not** ‘there is now a kettle/telephone in the mutual cognitive environment’ – the use of the definite article reflects their normal presence in what Fuchs called ‘household’ settings), but rather in depicting those referents as involved in

an event of a sort they are customarily involved in. This function I will call ‘event-central’ thetic sentences (an alternate term is Lambrecht’s ‘event-reporting’ sentences).

communicative functions of thetic sentences	
entity-central (presentational) (17) JOHN arrived. (18) The POLICE are coming.	event-central (event-reporting) (15) The TELEPHONE’s ringing. (16) The KETTLE’s boiling.

Again there is certainly room for overlap/gradation between the two types – while the sentences listed in this diagram are perhaps the most quintessentially ‘entity-central’ or ‘event-central’ of the examples considered above, many of the others may plausibly be construed in either type. Still, Sasse/Lambrecht’s two communicative functions seem to represent a crucial step in building an explanatory account of thetic sentences, one we will seek to push further in a dynamic processing analysis in Section 4.

2.3 Not quite ‘out of the blue’?

Although, as we have noted, thetic utterances are largely described in the literature as being uttered ‘out-of-the-blue’ (precisely because, as we have seen above, they are compatible with situations with no prior **verbalized** content), somewhat paradoxically they are usually presented in the literature together with appropriate context-setters intended to force a thetic reading. I have so far followed suit in this paper, in (1) as well as in all the ‘context:’ settings given in parenthesis before the examples in (15) - (30).

But note that these contexts may involve communication delivered verbally or by nonverbal means, through body-language or intonation – i.e. the boss’s tacit eyebrow-raising in (1), a surprised look to signal ‘what’s that noise’ prior to (15) or (16), a sigh of oh-no prior to (19) or (20), an aghast or forlorn expression prior to (21) or (22), the question Why are you so happy? prior to (23) or (24), a grimace prior to (25) or (26), the expectant glance of an attentive narrative-listener prior to (29) or (30), or more generally the question What happened? oft-used in the literature as a thetic-reading diagnostic. What all these contexts seem to share is that the thetic sentence is uttered in response to,

or even elicited by, a (verbal or nonverbal) expectation for an explanation of some state of affairs.

What role does this expectation play in the interpretation of thetic sentences? As for ‘entity-central’ thetic sentences such as (17) and (18), responding to such an expectation does not seem obligatory for their felicitous utterance. Namely, they might be uttered in response to a verbal or nonverbal question/expectation, e.g. What's that noise?, but they may also be uttered when no such expectation for an explanation has been communicated (picture a speaker looking out the window as the only person to see a car pull up, or one burglar warning another after a long period of silence). Consider again that these ‘entity-central’ thetic sentences describe little more than the appearance of the NP referent in the mutual cognitive environment, that the predicates arrived and are here do not contribute much beyond conventional modes of appearance. Moreover, both are ‘centered’ to the current, default situation as noted above by Fuchs. If the predicate is more descriptive about an unusual method of appearance/arrival, as in (31), or deals with a frame of reference removed from current concern, as in (32), a thetic reading is resisted, leaving only a narrow-focus reading:

(31) # JOHN's parachuted in.

(32) # JOHN's got there. (both # marks for thetic reading)

In other words, it seems that in ‘entity-central’ thetic sentences the mere mention of the new referent is enough to convey the brunt of the speaker's message: in (17), if the referent of John was previously inactive in the mutual cognitive environment, the mention of the stressed NP John... would suffice to lead the hearer to the conclusion that what is being communicated is an event involving John entering the mutual cognitive environment – the hearer might plausibly anticipate that this is likely by becoming physically present, likely in a conventional way. Indeed, this much would be conveyable if the speaker had chosen to utter just John! and left the rest to inference. We can therefore conclude that for ‘entity-central’ thetic sentences such as (17) and (18), if uttered in a context without a communicated expectation for an explanation as above, the predicate is rendered largely predictable at the mere mention of a new NP referent, because the predicate describes a conventional manner of appearance for such a referent and is centered to the default, current situation.

As for the ‘event-central’ thetic sentences including (15) and (16) (and to some extent many if not all of (19)-(30)), note that the mention of the new NP referent alone does not seem to suffice to render the predicate largely predictable in the same way. Rather, it is rendered predictable by the combination of the mention of the new NP referent plus an expectation for an explanation, expressed by verbal/nonverbal means.

In both cases, we are now drawing closer to understanding A&C’s insight that all thetic sentences ‘seem to draw the listener’s attention to a prominent NP’. Processing the subject NP bearing focal stress comes first in the utterance, in a situation involving previous verbal or nonverbal communication, gives the hearer a head start – not so much in anticipating precisely what the predicate will contain, but in anticipating the main implications of the utterance. For event-central thetic sentences this may be schematized as ‘something about X is the reason for Y’, for entity-central thetic sentences ‘X is now in the mutual cognitive environment’, where X is the subject NP and Y is some event requiring explanation.

It is one of the central claims of this paper, therefore, that the prevailing notion of thetic sentences being by definition appropriate when ‘uttered out of the blue’ is misleading and overlooks something key to a proper analysis of the thetic phenomenon (especially event-central thetic sentences). It is misleading because it confuses the necessarily discourse-new status of the subject NP and the predicate (i.e. neither being anaphorically linked to previous verbal material) as entailing the absence of any prior communication, verbal or nonverbal, bearing upon the interpretation. What it overlooks is that event-central (event-reporting) thetic sentences, in particular, presuppose a cause for an event requiring explanation. Quintessentially entity-central (presentational) thetic sentences such as ‘the POLICE are coming’, on the other hand, do seem to permit more of a pure ‘out of the blue’ context, i.e. when no expectation for an explanation has been communicated.

To sum up, Section 2 has shown that thetic sentences may be seen to vary along a certain cline, from more entity-central (for which the mention of the NP out of the blue may be enough to convey the brunt of the message) to more event-central (for which an expectation for an explanation plays a crucial part in interpretation).

3. Predictability and relevance

As the preceding sections of this paper were largely couched in informal terms, Section 3 will now work towards formalizing them to a somewhat greater extent by outlining and extending the relevance-theoretic approach to dynamic processing as it applies to thetic sentences, drawing a distinction between ‘weak’ and ‘strong’ predictability.

3.1 Pro-active focal stress

S&W (1995, 202-217) sketch out relevance-theory grounded, dynamic approach to processing, further discussed and advanced by Breheny (1996, 1998). This approach is centered around the idea, inspired by the literature on sentence parsing, that hearers posit successive anticipatory hypotheses about sentence content as each constituent is processed, with each such step guided by the Principle of Relevance. I will largely assume the reader's familiarity with Relevance Theory here, pointing out that our starting point is S&W's application of relevance-theoretic mechanisms below the whole-utterance level, which has not yet been extensively pursued in subsequent literature (although aside from Breheny's work see Kempson et al. 2001; Wedgwood 2005).

S&W (1995:204) write:

Because a sentence is produced and processed over time, the hearer will be in a position to access some of its constituent concepts, with their associated logical and encyclopedic entries, before others. For a speaker aiming at optimal relevance, efficient exploitation of this temporal sequencing will be crucial.

To illustrate this, we will first briefly walk through the processing of an utterance with final stress:

(context: What happened? / What was that noise?)

(33) John [broke the VASE_F].

On S&W's and Breheny's approach, upon processing of the first constituent in (33), the hearer will make hypotheses about how to resolve the referent of the subject NP, as well as about what to expect in terms of the structure/content of the remainder of the utterance. Here and in all such hypothesizing, the hearer is guided by the ‘relevance-

theoretic comprehension procedure' (S&W 2004:613): a processing heuristic which involves following a path of least effort, considering interpretative hypotheses in order of accessibility, stopping when expectations of relevance are satisfied. The hearer thus forms an anticipatory hypothesis that may be informally expressed as 'John (did something)'. Processing of the second constituent in (33), reaching John broke..., proceeds similarly, triggering a hypothesis about the intended meaning of broke among its various possible meanings, plus the positing of an anticipatory hypothesis 'John broke (something)'.

These early hypotheses (incomplete syntactic/semantic representations) are already subject to tentative attempts at pragmatic inferencing and may themselves have some cognitive effects (here plausibly the hypothesis 'John is to blame for the noise'), but because none of the processed constituents has borne focal stress, these cognitive effects are not presented by the speaker as necessarily relevant in their own right – S&W call such anticipatory hypotheses 'background hypotheses'; they may achieve relevance only by paving the way for and facilitating further processing (i.e. by reducing effort).

Reaching the stressed constituent, which in (33) is a VASE, represents a kind of watershed in the hearer's processing procedure. Namely, the speaker's use of focal stress indicates that the hearer is now expected to derive a hypothesis which is relevant in its own right, i.e. what S&W call a 'foreground hypothesis'. Since the stressed constituent is in this example sentence-final, the hearer will be in an especially convenient position for deriving such cognitive effects, having already processed every constituent in the full utterance. The use of sentence-final stress therefore may be seen as a way of presenting information step-by-step, leading the hearer through successive anticipatory hypotheses, each paving the way for the next and reducing effort – giving the hearer all the pieces of the puzzle, as it were, before the stressed constituent is reached.

Next consider the processing of an utterance with non-final focal stress, such as (34):

(context: What did John do to the vase?)

(34) John [BROKE _F] the vase.

Here the first step, the processing of John... proceeds as above, giving rise to the same background anticipatory hypothesis. But the second step of processing (34)

already reaches the stressed constituent, thus signaling that at this stage the hearer should be able to derive a foreground hypothesis about the content of the entire sentence, i.e. a hypothesis relevant in its own right.

Breheny (1996, 1998) calls non-final focal stress, as in (34), ‘pro-active’ – it prompts the hearer to seek an appropriate presupposition that can help in positing a foreground anticipatory hypothesis at a non-final stage in processing. As we have noted, this search for presuppositions follows the general relevance-theoretic comprehension heuristic. This heuristic plays a part in every stage of processing (not just the stage at which the stressed constituent is reached), but focal stress – both S&W and Breheny draw an analogy here to pointing at something physically – serves as a kind of ostensive stimulus on a sub-utterance level, drawing the hearer's attention to this stage of processing as being particularly worthy of an investment of processing effort, holding forth the promise of rich cognitive effects to be so derived. This means that at the stage at which non-final stressed constituent is reached, the hearer can expect the search for an appropriate presupposition to use in constructing an anticipatory hypothesis about the content of the whole utterance to be particularly fruitful. In our example (34), an appropriate presupposition can indeed be found at that stage, derived from the context question (this may be represented as ‘John (did something to) the vase’).

Under Breheny's ‘pro-active’ view of focus, therefore, sentences with non-final stress trigger greater processing effort earlier, forcing the hearer to spend more effort on constructing an anticipatory hypothesis of the whole sentence without yet having all the pieces of the puzzle. Wilson & Wharton (2006: 1567) sum up the relevance-theory view on stress marking as follows:

It follows from the Communicative Principle of Relevance that if two stress patterns differ in the amounts of processing effort required, the costlier pattern should be used more sparingly, and only in order to create extra, or different, effects.

In our examples, the effects of non-final stress in (34) differ from utterance-final stress in (33) in two ways: non-final stress urges the hearer to search for a presupposition of a particular form to use in processing, and this search is performed in the promise of yielding ‘foreground’ cognitive effects at an earlier stage.

Breheeny offers a generalization of this ‘pro-active’ impact of non-final focal stress: constituents which follow the stressed constituent will be ‘predictable’, i.e. accessible or derivable via bridging inferences⁹, **prior to** the stage of processing which involves the stressed constituent. That means that all presuppositions/referents necessary for correctly interpreting an utterance must become accessible not prior to processing of the utterance as a whole, as often implicitly assumed in the literature, but prior to the stage of processing when the stressed constituent is reached. One of the advantages of this approach is that it brings the notions of presupposition and referent activation down to the sub-utterance level. For instance, it helps provide an account of (35):

- (35) I want to learn to play the violin because I LIKE string instruments.
(Büring 2005)

Here string instruments may not be salient prior to the whole utterance, but is made accessible by encyclopedic knowledge triggered by violin (as a hyperonym) by the stage the stressed constituent LIKE is reached; thus Breheeny's generalization holds.

In Sax (2010), I have argued that this generalization, specifically Breheeny's definition of predictability, must be reformulated to account for certain sentences in English with non-final stress, especiallythetic sentences such as (9), repeated again:

- (context: What happened?)
(9) [My CAR broke down_F]

S&W note that in a thetic sentence like (9)

...there is a deep-rooted intuition [...] that the material following the subject noun phrase is predictable in some sense. The appropriate sense, we believe, is derivable from our notions of weak implicature and weak communication. (S&W 1995: 212)

⁹ Breheeny describes the bridging space or "B-space": "an antecedent is accessible in a B-space iff it is accessible in a P-space (for pronoun space) or it can be introduced via mutually manifest (encyclopedic) information associated with material in the P-space." On relevance and bridging cf. Gundel (1996)

The problem with Breheny's approach, pointed out in Sax (2010), is that on the thetic reading in (9) no presupposition of the form '(something) broke down' is accessible or derivable before the processing of the stressed subject NP, or at any processing stage for that matter. Instead, the predictability of broke down is crucially linked to the hearer's encyclopedic knowledge about cars in general – which S&W argue contains the “highly accessible and strongly confirmed assumptions that cars can break down, run out of petrol and fail to start, all of which would reasonably account for the speaker's failure to arrive on time”. (1995: 212)

These two quotations from S&W encapsulate the two crucial notions for the account to thetic utterances advocated in this paper: a weak kind of predictability and the hearer's expectation for an explanation. Each of these will now be considered in turn from the relevance-theoretic standpoint.

3.2 Weak vs. strong predictability

In (9) the post-focal material (broke down) becomes predictable not **prior to** the stage of processing at which the stressed constituent (my CAR) is reached, but rather **at** the very stage when the stressed constituent is reached (the referent of my car, after all, was itself not yet salient in the mutual cognitive environment until that point, and thus cannot have triggered the necessary encyclopedic knowledge at any prior stage). In Sax (2010) I proposed that this phenomenon be called ‘weak predictability’ and preliminarily explored some of its properties, suggesting that Breheny's notion of predictable material based on presuppositions should be recast as ‘strong predictability’ – this choice of weak/strong terminology is directly inspired by S&W's quote above.

The definitions presented in Sax (2010) were phrased to facilitate comparability to Breheny's generalization; here they are recast for simpler application to the current concern of thetic sentences:

- (36) Post-focal material is **strongly** predictable if it is accessible¹⁰ **prior to** processing of the focally stressed constituent.

¹⁰ Or accessible via bridging inferences in Breheny's "B-space" – see previous footnote.

- (37) Post-focal material is **weakly** predictable if it becomes accessible **upon** processing of the focally stressed constituent.

This means that a post-focal constituent which is strongly predictable must be anaphorically linked to previous linguistic material – i.e. to an accessible referent like the vase in (11) or an accessible presupposition like ‘(something) broke down’ in (12) – not to encyclopedic knowledge triggered by the NP bearing focal stress. A post-focal constituent that is weakly predictable, on the other hand, may become predictable via inferences drawing upon the encyclopedic knowledge triggered by the stressed NP.

We can again underscore the close link here to the relevance-theoretic concept of ‘strong vs. weak communication’ (cf. S&W, Carston 2002: 380-381): when post-focal material is strongly predictable, the accessible referent or presupposition sought by the hearer upon processing the constituent bearing non-final focal stress is often the single most accessible such referent or presupposition of the appropriate form in the mutual cognitive environment – for instance, having been rendered so by explicit representation in immediately prior discourse like in (11) or (12), repeated below. As we noted above, while positing anticipatory hypotheses the hearer follows the relevance-theoretic comprehension heuristic, entertaining hypotheses in order of accessibility; at the processing stage involving the constituent bearing non-final focal stress this procedure turns up the highly accessible referent the vase in (11) or highly accessible presupposition like ‘(something) broke down’ in (12), in either case directly attributable to previous linguistic material (here shown in boxed type). Indeed, this referent/presupposition may be so strongly accessible, and the related hypotheses so strongly communicated, as to license pronominalization as in (38) or ellipsis as in (39):

(context: What did John do to the vase?)

(11) John [BROKE_F] the vase.

(38) John BROKE it.

(context: What broke down?)

(12) [My CAR_F] broke down.

(39) My CAR!

By contrast, the post-focal material in (9) is weakly predictable, i.e. its predictability cannot be attributed to any material prior to the NP bearing focal stress itself:

(context: What happened?)

(9) [My CAR broke down_F]

(40) My CAR! / My damned CAR!

We can list four hallmark traits of a weakly predictable post-focal constituent, like the predicate broke down in (9), which result from the interaction of the ‘pro-active’ impact of non-final focal stress and the relevance-theoretic comprehension procedure:

A) The post-focal constituent spells out one of a set of weakly implicated¹¹ hypotheses posited at the processing stage when the constituent bearing non-final stress is reached, all of them more or less equally accessible (one possible such set for (2) is the set {break down; run out of petrol; fail to start}, suggested above by S&W); thus the relevance-theoretic comprehension heuristic followed by the hearer at that stage returns a set of hypotheses, rather than a single hypothesis as seen for (11) or (12) above.

B) One piece of evidence in favor of this analysis is that if syntactic considerations allow the post-focal constituent to undergo ellipsis, the same set will be weakly implicated by a fragmentary utterance consisting solely of stressed NP, as in (40) – a point made by S&W, and one we also made regarding the fragmentary John! above.

C) In both cases – either an explicit post-focal constituent as in (9) or a fragmentary utterance invoking inference as in (40) – this set of weakly implicated hypotheses is not rendered salient to the hearer prior to the stage in processing when the stressed constituent is reached, but rather triggered at that very stage, by the lexical semantics of the stressed constituent in conjunction with prior context (of course, the ‘out of the blue’ approach to thetic sentences such as (9) considers them analyzable in complete isolation from any prior context – but I discuss above and below the impact of the possibly nonverbal ‘expectation for an explanation’).

¹¹ A hypothesis is “weakly implicated if its recovery helps with the construction of a satisfactory interpretation, but is not essential because the utterance provides evidence for a wide array of roughly similar conclusions, any of which would do” (Wilson & Wharton 2006: 1569)

D) The post-focal constituent contributes little to the implicatures of the utterance as a whole (in S&W's terms: being 'background' rather than 'foreground') – it merely spells out or strengthens one of the set of weakly communicated hypotheses, whereas the main implicatures of the utterance as a whole are largely the same for all members of that set. For (9) this can be represented as 'something about my car is the reason why I am late' – this implicature holds regardless of which member of the set of weak hypotheses is actually asserted in (9) or left to inference in (40).

Weak predictability is not unique tothetic utterances in English, but is also seen in other cases of non-final focal stress like the relative infinitive construction in:

(Next month we may be out on the street...)

(41) I'm [looking for a HOUSE to rent_F]. (Bolinger 1972, one speaker)

Here, to rent is weakly predictable and exhibits the same four traits described above: A) it spells out one of a set of weakly implicated hypotheses, e.g. plausibly {to rent, to buy, to squat in}, B) this set is weakly communicated by the truncated 'I'm looking for a HOUSE' in the same context, C) this set is derived from encyclopedic knowledge not rendered accessible until house is processed, D) which particular member of that set is actually ultimately uttered is presented by the speaker as largely immaterial to the main implicatures of the utterance, e.g. possibly including 'I have something to worry about' / 'I'll be busy soon' / 'Perhaps you can help.'

Weak predictability does not seem to play a role in the interpretation of non-final focal stress in all languages (see Ladd 1996), or at least not to the same extent as in English. But speakers of English can indeed exploit weak predictability in situations such as (1)/(2) and (42), choosing to put 'background' before 'foreground' in S&W's terms and strategically exploit a tradeoff in earlier processing effort vs. earlier cognitive effects – and by the familiar relevance-theoretic logic (grounded in the Presumption of Optimal Relevance), since they **can** exploit weak predictability, hearers will **expect** them to.

This leads us back to the observation that there is a timing factor involved in the strategy the speaker adopts in guiding the hearer's search for optimal relevance:

the sooner disambiguation and reference assignment are achieved, the less processing effort will be required. The greater the number of possible interpretations that has to be borne in mind as the utterance proceeds, the greater the processing effort. It follows that a speaker aiming at optimal relevance should phrase her utterance so as to facilitate early – and correct – disambiguation. (S&W 1995: 204).

3.3 ‘Expectation for an explanation’ as a processing constraint

What role does what we have called an ‘expectation for an expectation’ play in this relevance-theoretic account of the speaker's utterance of and hearer's processing of a thetic sentence? We saw above that such an expectation is very commonly communicated by the context in which a thetic sentence is uttered (especially of the ‘event-central’ variety), either non-verbally as in (1)-(2), or verbally such as by the context question used in (9) What happened?, frequently used in the literature to force a thetic reading. Such an expectation may be viewed as a kind of presupposition: presupposing that there is some causing-event which is the cause of the caused-event needing explanation. The causing-event may be conceived of as a presupposed empty event, at first entirely filled out by place-holding variables but standing in a causal relation to the caused-event.

In the classic late-for-work example in (2)/(9) above, once such an expectation on the part of the boss becomes mutually manifest, under the Principle of Optimal Relevance the boss will expect the employee's response to attain relevance in respect to that expectation, and moreover the employee will expect her response to be processed by the boss in respect to that expectation, etc. The presupposition that this expectation gives rise to, an empty-but-causally-linked presupposed event, is rendered highly salient in the context by the communicative situation, and as such it will be considered at an early point in the hearer's search for an appropriate anticipatory hypothesis upon processing my CAR... bearing focal stress. The hearer will combine this presupposition with a referent assigned to my car to derive an anticipatory hypothesis that may be represented ‘my car (did something) and that is why I'm late’.

By triggering this causally embedded hypothesis, an expectation for an explanation serves as a constraint in the hearer's application of the relevance-theoretic comprehension heuristic. Namely, it constrains how the encyclopedic knowledge

triggered by the subject NP in a thetic sentence is brought to bear in the processing procedure: the hearer will only entertain those (weak) hypotheses about the predicate which offer plausible explanations for the caused-event, as only those hypotheses can be expected to satisfy the hearer's expectations of relevance. For instance, the set of weakly implicated hypotheses S&W suggest for the classic faulty-car example, {break down; run out of petrol; fail to start}, is by no means a full set of predicates that may be triggered by the lexical semantics of car but is constrained only to those predicates which offer plausible explanations for the caused-event, e.g. being late.

The same observation applies to all the thetic utterances analyzed above as 'event-central': in response to the verbal/nonverbal expectation for an explanation What's that noise?, the hearer's processing procedure upon encountering the initial stressed NP in The TELEPHONE's ringing/The KETTLE's boiling will entertain not the full set of predicates possibly triggered by the NP's lexical semantics, but only those predicates that reflect ways in which they plausibly and conventionally make noise. Again, this is confirmed by what may be weakly communicated by the truncated responses The TELEPHONE/The KETTLE.

4. Characteristics and constraints revisited

Sections 1-3 have reviewed various characteristics of thetic utterances, then proposed two elements of a new analysis of thetic utterances in a Relevance Theory framework: highlighting the shortcomings of the frequent 'out of the blue' approach (recognizing instead the impact of a verbally/nonverbally communicated expectation for an explanation) and introducing a distinction between weak vs. strong predictability. With these tools in hand, we can now reexamine the characteristics, traits, and constraints of thetic sentences, striving to show briefly how this analysis dovetails with the intuitions of previous accounts yet improves upon them in significant ways.

4.1 One step, not two

To begin with, our analysis can be seen to shed further light on A&C's observation cited above, that a common element which thetic sentences share is that they "seem to draw

the listener's attention to a prominent NP, saying 'look at NP'". The focal stress on the subject NP – by S&W and Breheny's analogy to physical pointing – may be said to draw the hearer's attention to that NP in that it urges the hearer to invest greater effort in processing at that stage, in the promise of richer cognitive effects.

This analysis moreover captures the intuitions noted in the literature of how processing of thetic sentences proceeds in 'one fell swoop' (Jacobs) or one step rather than two, that there is a kind of 'communicative fusion' (Sasse), 'conceptual unity' (Chafe) or 'integration' between subject and predicate. By encouraging greater effort at an early stage, the focal stress on the subject NP signals that a significant share of the cognitive effects of the utterance can be derived from processing of the subject NP itself, abstracting over the weakly implicated set of anticipatory hypotheses turned up by the relevance-theoretic comprehension procedure at that stage. When processing of the predicate – which in S&W's terms represents 'background' material coming after 'foreground' – does ultimately proceed, it achieves little more than confirming one member of that already hypothesized set, triggering a much smaller share of the cognitive effects of the utterance as a whole. This is the source of the intuition of processing in one step rather than two.

Next, we have seen that the close relationship which must obtain between subject and predicate for a thetic reading to be supported is largely rooted in lexical semantics, yet mediated by context and convention in complex ways – this is represented straightforwardly in our approach: encyclopedic knowledge triggered by the new, stressed subject NP is the source of weakly implicated hypotheses about the predicate, but the set of such hypotheses actually considered by the hearer is constrained by the hearer's search for relevance in the mutual cognitive environment plus the possible further constraint imposed by an expectation for an explanation.

As S&W state, with their approach to thetic sentences, which we have further expanded upon in this paper, 'it is possible to do rather better' than analyses relying upon comparing the 'relative semantic weight' (e.g. Bolinger 1972) or relative 'newsworthiness' (e.g. A&C) of subject and predicate, to which list we can moreover add the 'communicative dynamism' of the Prague school tradition (e.g. Firbas 1992): all these approaches have ultimately failed to rigorously define metric criteria by which the weight of one constituent or lexical item is gauged against that of another, by either the

speaker or hearer. By drawing upon the framework of Relevance Theory, which brings with it the independently motivated and cognitively-grounded metric notion of relevance (framed as a tradeoff of effort vs. cognitive effects) as well as an advanced model of speaker-hearer interaction, the approach to thetic sentences outlined here is able to be more rigorous. Rather than appealing to intuitive notions of relative weight, we can claim instead that a speaker deciding where to place focal stress compares the cognitive effects which the hearer may be expected to derive at one point (or constituent) in processing of an utterance, against the effects the hearer can be expected to derive at another point (constituent), all the while taking account of the fact that the hearer will derive anticipatory hypotheses following the relevance-theoretic comprehension heuristic.

4.2 New or old?

We noted above that the challenge thetic sentences have always posed to theories of information structure which rely upon bipartite divisions of sentence constituents into two disjoint categories (new/old, topic/focus, rheme/theme), posited on the whole utterance level. We can briefly recap this by saying that the predicate in a thetic sentence represents new information, yet by going unstressed behaves as if it were already old. This challenge has been handled in various ways in the literature, such as by positing special stress marking rules (Selkirk 1984), assuming implicit spatio-temporal ‘stage’ topics (Erteschik-Shir 1997), or considering thetic sentences to represent a construction interpreted globally through ‘paradigmatic contrast’ (Lambrecht 1994, 2000). The approach to thetic sentences outlined here, by drawing upon S&W’s and Breheny’s insights which bring inferential pragmatic reasoning down to below the whole-utterance level, takes a more promising tack: predicates such as ...is boiling or ...are coming may be entirely new and unpredictable prior the processing of a thetic utterance as a whole, yet become predictable (to some stronger or weaker degree) at an intermediate stage in processing of the utterance, namely upon processing of initial NPs the KETTLE... or the POLICE... bearing focal stress, respectively.

As noted above, Sasse concludes that the thetic/categorical distinction is “not a matter of information structure” as often assumed, suggesting that it can explained in

terms of “expectations” which are “only a partially a matter of information structure” (Sasse 1987: 523); this notion of thetic sentences reflecting an ‘expectation’ on the part of the hearer goes back at least to Schmerling (1976). I have taken this further, showing that an expectation for an explanation, whether expressed verbally or nonverbally, may crucially constrain the hearer's processing of a thetic sentence (especially important for ‘event-central’ thetic sentences).

4.3 Constraints on form as constraints on information

Pressing ahead now with the ‘vexing’ question raised above of which predicates are able to support thetic readings in a subject-stressed sentence, our approach here can offer some justification for the syntactic and lexical constraints on thetic sentences outlined in Section 1.4. Turning first to the lexical constraints, the observation that the predicate in a thetic sentence must be ‘centered’ to the default, current situation finds a natural explanation in terms of accessibility: aspects of the current situation are generally more accessible to the hearer in the mutual cognitive environment than situations which are not of current concern. Following the relevance-theoretic comprehension heuristic, the hearer will naturally consider possible hypotheses in order of accessibility, and will thus first posit hypotheses about predicates dealing with the situation of current concern. The speaker, recognizing this, will only use thetic stress marking when the hearer can be expected to make the right kind of hypotheses following the relevance-theoretic comprehension heuristic, thus generally when the predicate bears upon the situation of current concern.

Next, we can say that the subject and predicate (or more specifically the verb) of a thetic sentence must each be fully lexically expressed (i.e. not pronominal or anaphoric) each for similar but different reasons. A thetic sentence by definition introduces a new NP referent and a new predicate about that referent, whereas a pronominal/anaphoric NP requires a highly accessible referent which is therefore not new. When a pronominal subject NP bears focal stress, as in the example HE arrived, the subject NP is interpreted as picking one referent out of a set of accessible referents and does not render accessible encyclopedic knowledge about a new NP referent (which we saw above is a key element in the mechanism of weak predictability); thus this example supports only a narrow-

focus reading (i.e. the predicate must be strongly predictable, already accessible in the form of a presupposition). The predicate in a thetic utterance, in turn, must be fully lexically expressed because, as we have seen, processing of the predicate serves to confirm one of a set of more or less equally accessible, weakly predictable anticipatory hypotheses about the predicate posited by the hearer. Use of an anaphoric predicate such as in JOHN did would indicate that there is a single very highly accessible predicate to which it relates; hence again this example only supports a narrow-focus reading (i.e. the predicate must again be strongly predictable).

Turning next to more syntactic considerations, the tendency for intransitive/passive predicates in thetic sentences can again be seen as a consequence of the mechanism of weak predictability: the set of weakly predictable anticipatory hypotheses which the hearer derives upon processing the stressed subject NP is largely based upon encyclopedic knowledge about the newly introduced referent; such knowledge suggests hypotheses about possible events/predicates that the referent may be involved in given the situation of current concern, but not about which other new referents might also be involved in those events/predicates. This implies a general restriction against introducing referents 'by the back door' as it were: the example JOHN broke the vase resists a thetic reading because expecting the hearer to derive both a new predicate and another referent to which that predicate applies from encyclopedic knowledge about the referent of John, all at the stage at which the stressed subject NP is processed, would make extraordinarily high demands in terms of processing effort at that stage. Therefore, this example easily supports only a narrow-focus reading (i.e. the predicate is yet again interpreted as strongly predictable).

On the other hand, we saw above that some transitives can bear a thetic reading and that objects then show the reverse tendency to subjects and verbs, strongly tending to be pronominal in thetic sentences rather than lexically expressed. This can be explained by noting that referents of object pronouns must already be highly accessible in the mutual cognitive environment, thus their incorporation as additional participants in predicates /events hypothesized based on encyclopedic knowledge triggered by the subject NP plausibly requires less effort from the hearer than for non-pronominal objects.

Recall that Fuchs points to the underlying principle of the syntactic and lexical restrictions on thetic sentences as being "a restriction on the amount of information

conveyable" (Fuchs 1980: 455), which we rephrased as a cognitive restriction on the amount of information that can be processed by the hearer in Jacobs' 'one fell swoop'. This subsection has sought to show how such a cognitive restriction naturally falls out of the mechanism of weak predictability and the relevance-theoretic comprehension heuristic.

4.4 Predictability revisited

By distinguishing between strong and weak predictability, this analysis also offers a way of responding to various criticisms against the idea that the unstressed predicate in a thetic sentence is predictable. For instance, consider the following putative counterexamples put forward by Schmerling (1976: 41), in which for her "the verb does not seem at all predictable":

- (42) I have to go home – my COUSIN's coming.
- (43) Hey – your COAT's on fire!
- (44) Watch out – there's a CAR coming.
- (45) Waiter, there's a FLY in my soup.

Schmerling (1976: 41, boxing added)

Firstly, note that Schmerling is obviously **right** in one sense: in each of these examples, the predicate is not strongly predictable – i.e. it is not anaphorically linked to previous discourse, derivable through the application of a presupposition accessible by the stage the stressed subject NP is processed, as the single most accessible anticipatory hypothesis turned up by the relevance-theoretic processing heuristic.

But secondly, note that all four of these thetic sentences – putatively uttered 'out of the blue' – are cited by Schmerling together with previously verbalized material (shown in boxed type) setting up what I have called an 'expectation for an explanation'. We have seen above how in each case, at the stage focal stress is processed, the combination of this expectation plus the encyclopedic knowledge triggered by the subject NP (partial output of processing at this stage may be represented as 'some event dealing with my

cousin is why I have to go home’, ‘something about your coat is why I am screaming to get your attention’, ‘something about a fly is why I called you over’, etc.) leads the hearer to posit a constrained set of weakly predictable hypotheses, i.e. plausible predicates involving the cousin, coat, or fly in question. Thus, Schmerling is **wrong** in the sense that at this stage the predicate in each of these examples is indeed to some extent predictable, becoming weakly predictable when the stressed subject is processed.

Moreover, each of Schmerling's examples illustrates how the main cognitive effects ofthetic sentences are derivable at the point when the stressed NP is reached (that the hearer should look at her coat / stop crossing the street / in some way tend to the table, etc.), largely abstracting over which specific member of the weakly implicated set of hypotheses is in fact uttered (or not uttered – again consider the similar effect of all the above NPs if uttered in isolation in the same contexts: e.g. Hey, your COAT!).

4.5 Form and motivation

The preceding discussion raised two questions concerning the motivation behind the $\acute{S}P$ (stressed subject / unstressed predicate) form ofthetic sentences. We considered Lambrecht's ‘paradigmatic contrast’ argument that the purpose of $\acute{S}P$ stress marking onthetic sentences to maximally differentiate them from the $S\acute{P}$ stress marking on categorical sentences, whereas the shared $\acute{S}P$ stress marking in thethetic and narrow-focus readings of subject-stressed sentences is a by-product of thisthetic/categorical contrast. The relevance-theoretic approach outlined in this paper instead sees the $\acute{S}P$ stress marking as itself **driving** the hearer's process of interpretation on the sub-utterance level, by encouraging greater effort at the stage the stressed subject is reached. This has several advantages over Lambrecht's arguments: it explains many of the special properties ofthetic predicates, it does not require a hearer to invoke systemic linguistic knowledge about a broaderthetic/categorical "paradigmatic contrast" when recognizing and interpreting athetic sentence, and lastly it suggests a natural motivation for the shared $\acute{S}P$ stress marking in thethetic and narrow-focus readings of subject-stressed sentences (hearer processing proceeds along similar lines in both, driven by the $\acute{S}P$ stress marking, with the difference being in terms of predictability: the predicate is strongly predictable in a narrow-focus reading, weakly predictable in thethetic reading).

4.6 Classification revisited

We can now briefly reconsider the types into which we classified thetic sentences above, in terms of the mechanism of weak predictability.

A&C's 'verbs of (dis)appearance' class of thetic sentences (examples (17)-(20) above) has largely been dealt with in the discussion of 'entity-central' sentences – they mainly serve to introduce or eliminate a new referent, whereby the manner of the appearance or disappearance is weakly predictable and centered to the situation of current concern. A&C's 'empty verb' class (examples (15)-(16) above), in turn, consists of thetic sentences used in situations where encyclopedic knowledge triggered by the NP raises a relatively narrow set of weakly predictable hypotheses (hence A&C's observations above that a kettle 'is capable of few activities other than boiling' in 'household' situations). Thus for both these two classes, the thetic sentence may or may not be used in response to any expectation for an explanation, as mention of the NP alone may or may not suffice to trigger a sufficiently constrained set of weak hypotheses.

The next class, A&C's 'misfortune' class which we have followed other authors in expanding to include 'fortune' (examples (21)-(24) above), seems to rest more crucially on such an expectation, namely an explanation for some indication of good/bad fortune (possibly intonation, facial expression, wow or other more explicit verbal material).

Sasse's 'body parts' class (examples (25)(26) above) can be analyzed along lines similar to both the 'empty verb' and '(mis)fortune' classes: body parts are likely to be relevant only in connection with problems they are causing, e.g. to explain a painful grimace, limp, or other possibly nonverbal behavior. Finally, Sasse's classes of weather expressions and scene-setting descriptions (examples (27)-(30) above) are tractable as responding to a standing expectation for further information about the situation of common concern, i.e. the physical environment, a narrative literary setting, etc.

To conclude, all the classes of thetic sentences, ranging on a cline from entity-central to event-central (Sasse's/Lambrecht's communicative functions), are simultaneously seen to range in terms of how crucial an expectation for an explanation of a certain state of affairs is for the hearer to derive a sufficiently constrained, weakly predictable set of

anticipatory hypotheses. The impact of the mechanism of weak predictability is therefore traceable in all the proposed classes.

5. Conclusions

After first defining thetic (all-new, subject-stressed) sentences and reviewing the properties/classes of thetic sentences reported in the literature, this paper has outlined a dynamic, relevance-theoretic approach to thetic sentences in English by building upon the approaches to focal stress put forward in S&W and Breheny (1996, 1998). I have sought to show that the ‘out of the blue’ approach to thetic sentences widely assumed in the information-structure literature is misleading, and that the properties of thetic utterances can be better explained in a relevance-theoretic framework in terms of ‘pro-active’ non-final stress on the subject NP and the phenomenon of weak (vs. strong) predictability, often constrained by a verbal or non-verbal expectation for an explanation of a certain state of affairs. I have argued that this relevance-theoretic approach improves upon previous analyses of the intuition that thetic sentences are processed in one step rather than two, constraints on the lexical and syntactic form of thetic sentences, the motivation for the $\acute{S}P$ stress marking on thetic utterances, and other aspects.

On a broader level, S&W comment that thetic sentences raise the question of whether focal stress marking is sometimes subject to arbitrary, purely linguistic constraints or whether it more generally follows "in some interesting way from pragmatic principles" (S&W 1995: 211-212). By looking at the impact of the relevance-theoretic comprehension procedure on sentence processing below the whole-sentence level, this paper has tried to lend further weight to the latter, ‘highlighting’ position: the special focal stress marking in thetic sentences is argued to follow in a principled way from independently-motivated pragmatic principles of wider application.

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